



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2023 Year 12 Course Assessment Task 3

Wednesday, 31 May 2023

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- A NESA Reference Sheet is provided.
- Attempt **all** questions.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Commence each new question on a new booklet. Write on both sides of the paper.
- At the conclusion of the examination, arrange the booklets used in the correct order after this paper and hand to examination supervisors.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Mr Ho

☐ 12MXX.2 – Mr Sekaran

☐ 12MXX.3 – Ms Ham

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{13}$	$\overline{8}$	$\overline{15}$	$\overline{36}$	

Question 1 (13 Marks)		Commence a NEW writing booklet	Marks
(a)	i. Find the real numbers A , B and C such that		3
		$\frac{28}{(5+x^2)(3-x)} = \frac{Ax+B}{5+x^2} + \frac{C}{3-x}$	
	ii. Hence find: $\int \frac{28}{(5+x^2)(3-x)} dx$.		3
(b)	Evaluate as an exact value: $\int_0^{\frac{1}{6}} \frac{\ln(1-3x)}{(1-3x)^3} dx$.		3
(c)	Use a suitable substitution to evaluate as an exact value:		4
		$\int_4^8 \frac{1}{x^3 \sqrt{x^2-16}} dx$	

Question 2 (8 Marks)		Commence a NEW writing booklet	Marks
(a)	i. Show that, for $x > 1$,		1
		$0 < \frac{1}{x\sqrt{x}} < \frac{1}{x} < \frac{1}{\sqrt[3]{x^2}}$	
	ii. Hence show that, for $t \geq 1$,		2
		$\frac{2(\sqrt{t}-1)}{\sqrt{t}} \leq \ln t \leq 3(\sqrt[3]{t}-1)$	
	iii. Hence find $\lim_{t \rightarrow \infty} \frac{\ln t}{\sqrt{t}}$.		2
(b)	Consider the function $f(x) = \tan^{-1} x$.		
	i. Show that $f''(x) < 0$ for $x > 0$.		1
	ii. Hence by considering the concavity of $f(x)$, or otherwise, show that for $a > b > 0$:		2
		$\tan^{-1}(a-b) + \tan^{-1}(a+b) < 2 \tan^{-1} a$	

Question 3 (15 Marks)

Commence a NEW writing booklet

Marks

- (a) The line ℓ_1 passes through the point $(0, 2, 1)$ and is parallel to the vector $\underline{p} = -\underline{i} - 4\underline{j} + 9\underline{k}$.

i. Find $\underline{r}$, the vector equation of ℓ_1 , in terms of $\lambda \in \mathbb{R}$.

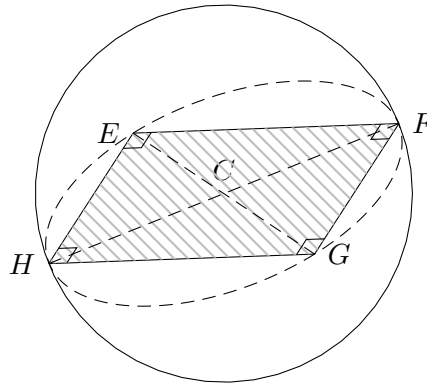
1

Let $\underline{s} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 1 \\ 5 \end{pmatrix}$ be the vector equation of the line ℓ_2 , where $\mu \in \mathbb{R}$.

ii. Prove that ℓ_1 and ℓ_2 are skew.

4

- (b) The sphere S has its centre at point C . $EFGH$ is a square such that E , F , G and H are all points on the sphere S , and its diagonals intersect at C . The points $E(0, 3, 4)$, $F(-4, 12, 3)$ and $G(4, 15, -2)$ are given.



Let O be the origin.

i. Prove that $\overrightarrow{OC} = \frac{1}{2} (\overrightarrow{OE} + \overrightarrow{OG})$ and show that C has coordinates $(2, 9, 1)$.

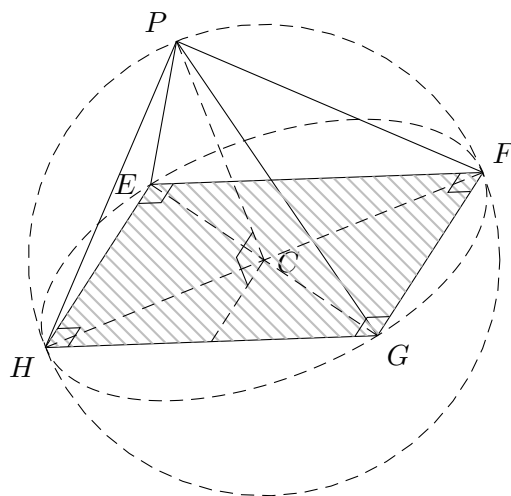
2

ii. Hence find the vector equation of the sphere S . Let $\underline{r}$ be the position vector of any point on the surface of the sphere.

2

Examination continues overleaf...

$P(a, b, c)$ is a point on the sphere S such that $PEFGH$ is a right square pyramid inscribed in the sphere. PC is the perpendicular height of $PEFGH$, as shown in the diagram.



- iii. Using vector methods, show that: 2

$$-2a - 6b + 3c = -55$$

$$-6a + 3b + 2c = 17$$

- iv. Hence find all possible coordinates for point P . 4

End of paper.

Suggested Band E4 Responses

Question 1 (Sekaran & Ham)

- (a) i. ✓ [1] for one correct value of A , B or C
 ✓ [1] for sufficient working out
 ✓ [1] for all correct values of A , B or C

$$\frac{28}{(5+x^2)(3-x)} = \frac{Ax+B}{5+x^2} + \frac{C}{3-x}$$

$$28 = (Ax+B)(3-x) + C(5+x^2)$$

Let $x = 3$:	$28 = 14C$	$\Rightarrow$	$C = 2$
Let $x = 0$:	$28 = 3B + 10$	$\Rightarrow$	$B = 6$
Let $x = 1$:	$28 = 2(A + 6) + 12$	$\Rightarrow$	$A = 2$

$\therefore A = 2$, $B = 6$ and $C = 2$.

- ii. ✓ [1] for correct $\ln(5+x^2)$ integral
 ✓ [1] for correct $\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)$ integral
 ✓ [1] for correct $\ln|3-x|$ integral with absolute value

$$\begin{aligned} \int \frac{28}{(5+x^2)(3-x)} dx &= \int \frac{2x+6}{5+x^2} + \frac{2}{3-x} dx \\ &= \int \frac{2x}{5+x^2} + \frac{6}{5+x^2} + \frac{2}{3-x} dx \\ &= \ln(5+x^2) + \frac{6}{\sqrt{5}} \tan^{-1}\left(\frac{x}{\sqrt{5}}\right) - 2 \ln|3-x| + C \quad \checkmark\checkmark\checkmark \end{aligned}$$

- (b) ✓ [1] for correct integration by parts
 ✓ [1] for correct subsequent integral
 ✓ [1] for correct exact value

$$\begin{aligned} \text{Let } u &= \ln(1-3x) \quad \rightarrow \quad u' = \frac{-3}{1-3x} \\ \text{and } v' &= \frac{1}{(1-3x)^3} \quad \rightarrow \quad v = \frac{1}{6(1-3x)^2} \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{1}{6}} \frac{\ln(1-3x)}{(1-3x)^3} dx &= \left[\frac{\ln(1-3x)}{6(1-3x)^2} \right]_0^{\frac{1}{6}} - \int_0^{\frac{1}{6}} \frac{-1}{2(1-3x)^3} dx \quad \checkmark \\ &= \frac{1}{6} \left(4 \ln\left(\frac{1}{2}\right) - 0 \right) + \frac{1}{2} \left[\frac{1}{6(1-3x)^2} \right]_0^{\frac{1}{6}} \quad \checkmark \\ &= -\frac{2}{3} \ln 2 + \frac{1}{12} (4-1) \quad \checkmark \\ &= \frac{1}{4} - \frac{2}{3} \ln 2 \end{aligned}$$

- (c) ✓ [1] for correct substitution with bounds for integral in terms of θ
 ✓ [1] for correctly simplifying integral using substitution
 ✓ [1] for correct primitive
 ✓ [1] for correct exact value

Let $x = 4 \sec \theta$.

$$\begin{aligned}
 & \int_4^8 \frac{1}{x^3 \sqrt{x^2 - 16}} dx \\
 &= \int_0^{\frac{\pi}{3}} \frac{4 \sec \theta \tan \theta}{64 \sec^3 \theta \sqrt{16 \sec^2 \theta - 16}} d\theta \quad \checkmark \\
 &= \int_0^{\frac{\pi}{3}} \frac{\tan \theta}{64 \sec^2 \theta \tan \theta} d\theta \\
 &= \frac{1}{64} \int_0^{\frac{\pi}{3}} \cos^2 \theta d\theta \quad \checkmark \\
 &= \frac{1}{128} \int_0^{\frac{\pi}{3}} 1 + \cos 2\theta d\theta \\
 &= \frac{1}{128} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}} \quad \checkmark \\
 &= \frac{1}{128} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{4} - 0 \right) \\
 &= \frac{1}{128} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{4} \right) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \frac{dx}{d\theta} &= -\frac{(-4 \sin \theta)}{\cos^2 \theta} \\
 dx &= 4 \sec \theta \tan \theta d\theta
 \end{aligned}$$

$$\theta = \cos^{-1} \left(\frac{4}{x} \right)$$

$$\begin{aligned}
 x = 8 &\rightarrow \theta = \frac{\pi}{3} \\
 x = 4 &\rightarrow \theta = 0
 \end{aligned}$$

Question 2 (Ham)

- (a) i. ✓ [1] for correct proof with sufficient working

$$\begin{aligned}
 x^4 &< x^6 < x^9 && \text{for } x > 1 \\
 \sqrt[3]{x^2} &< x < \sqrt{x^3} \\
 \frac{1}{\sqrt{x^3}} &< \frac{1}{x} < \frac{1}{\sqrt[3]{x^2}} && (x > 1) \\
 \therefore 0 &< \frac{1}{\sqrt{x^3}} < \frac{1}{x} < \frac{1}{\sqrt[3]{x^2}} && \text{as } x > 1 \quad \checkmark
 \end{aligned}$$

- ii. ✓ [1] for correct inequality (or equal) for primitives from (i)
 ✓ [1] for final result with correct working

Note that for $t = 1$, all integrals equal 0:

$$\int_1^1 \frac{1}{\sqrt{x^3}} dx = \int_1^1 \frac{1}{x} dx = \int_1^1 \frac{1}{\sqrt[3]{x^2}} dx = 0$$

So using part (i):

$$\begin{aligned}
 \int_1^t \frac{1}{\sqrt{x^3}} dx &\leq \int_1^t \frac{1}{x} dx \leq \int_1^t \frac{1}{\sqrt[3]{x^2}} dx \\
 -2 \left[\frac{1}{\sqrt{x}} \right]_1^t &\leq [\ln x]_1^t \leq 3 \left[\sqrt[3]{x} \right]_1^t \quad \checkmark \\
 -2 \left(\frac{1}{\sqrt{t}} - 1 \right) &\leq \ln t \leq 3 \left(\sqrt[3]{t} - 1 \right) \\
 \therefore \frac{2(\sqrt{t} - 1)}{\sqrt{t}} &\leq \ln t \leq 3 \left(\sqrt[3]{t} - 1 \right) \quad \checkmark
 \end{aligned}$$

- iii. ✓ [1] for correct limit of $t \rightarrow \infty$ for both bounds
 ✓ [1] for correct limit with justification

Multiplying by $\frac{1}{\sqrt{t}}$ from (ii): (note: $\sqrt{t} > 0$)

$$\frac{2}{\sqrt{t}} - \frac{2}{t} \leq \frac{\ln t}{\sqrt{t}} \leq \frac{3}{\sqrt[6]{x}} - \frac{3}{\sqrt{t}}$$

$$\begin{aligned}
 \lim_{t \rightarrow \infty} \left(\frac{2}{\sqrt{t}} - \frac{2}{t} \right) &= 0 \\
 \lim_{t \rightarrow \infty} \left(\frac{3}{\sqrt[6]{x}} - \frac{3}{\sqrt{t}} \right) &= 0 \quad \checkmark
 \end{aligned}$$

As $t \rightarrow \infty$, $0 \leq \frac{\ln t}{\sqrt{t}} \leq 0$.

$$\therefore \lim_{t \rightarrow \infty} \frac{\ln t}{\sqrt{t}} = 0 \quad \checkmark$$

- (b) i. ✓ [1] for correct proof with sufficient working

$$f'(x) = \frac{1}{1+x^2}$$

$$f''(x) = \frac{-2x}{(1+x^2)^2}$$

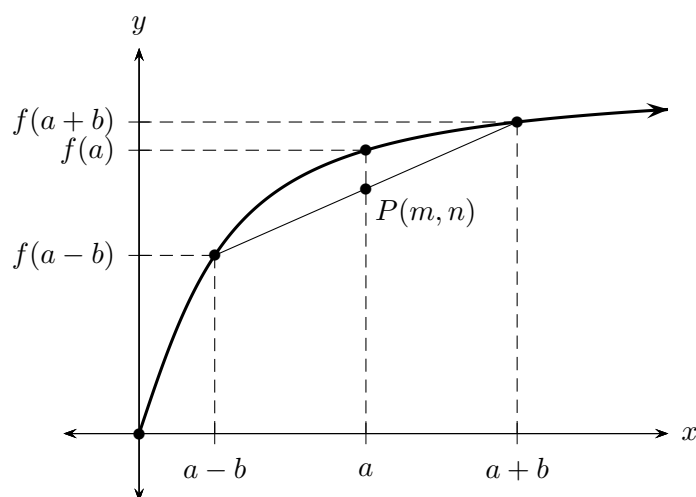
Since $(1+x^2)^2 > 0$ for $x > 0$,

$$\therefore f''(x) > 0 \text{ for } x > 0 \quad \checkmark$$

- ii. ✓ [1] for establishing inequality with sufficient explanation
 ✓ [1] for correct proof with sufficient working

From part (i), $f(x)$ is concave down and increasing for $x > 0$.

Let $P(m, n)$ be the midpoint of the secant:



Note: diagram is not required.

$$n < f(a) \quad \checkmark$$

$$\frac{\tan^{-1}(a-b) + \tan^{-1}(a+b)}{2} < \tan^{-1}(a)$$

$$\therefore \tan^{-1}(a-b) + \tan^{-1}(a+b) < 2 \tan^{-1}(a) \quad \checkmark$$

Question 3 (Ho)

- (a) i. ✓ [1] for correct answer

$$\underline{r} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -4 \\ 9 \end{pmatrix} \quad \checkmark$$

- ii. ✓ [1] for establishing correct set of equations from parametric forms
 ✓ [1] for proving that $\underline{r}$ and $\underline{s}$ do not intersect with correct conclusion
 ✓ [1] for proving that $\underline{r}$ and $\underline{s}$ are not parallel with correct conclusion
 ✓ [1] for proving that ℓ_1 and ℓ_2 are skew with sufficient justification throughout

If ℓ_1 and ℓ_2 intersect, then $\underline{r} = \underline{s}$:

$$\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -4 \\ 9 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 1 \\ 5 \end{pmatrix}$$

From the parametric equations:

$$\lambda + 5\mu = -3 \quad (3.1)$$

$$4\lambda + \mu = 4 \quad (3.2)$$

$$-9\lambda + 5\mu = 3 \quad \checkmark \quad (3.3)$$

Equation (3.1) – (3.3):

$$10\lambda = -6 \quad \Rightarrow \quad \lambda = -\frac{3}{5}$$

Equation (3.1) – $5 \times$ (3.2):

$$-19\lambda = 17 \quad \Rightarrow \quad \lambda = -\frac{17}{19}$$

$\therefore$ $\underline{r}$ and $\underline{s}$ do not intersect (no such $\lambda \in \mathbb{R}$ for which $\underline{r} = \underline{s}$). $\checkmark$

Considering the direction vectors for $\underline{r}$ and $\underline{s}$:

$$\begin{pmatrix} -1 \\ -4 \\ 9 \end{pmatrix} \neq \psi \begin{pmatrix} 5 \\ 1 \\ 5 \end{pmatrix}$$

$\therefore$ $\underline{r}$ and $\underline{s}$ are not parallel (no such $\psi \in \mathbb{R}$ for which their direction vectors are scalar multiples of each other). $\checkmark$

Hence ℓ_1 and ℓ_2 are skew. $\checkmark$

- (b) i. ✓ [1] for correct proof of $\overrightarrow{OC}$ sufficient working
 ✓ [1] for showing coordinates of C with sufficient working and conclusion

Diagonals of a square bisect each other.

$$\begin{aligned}\overrightarrow{EC} &= \frac{1}{2}\overrightarrow{EG} \\ \overrightarrow{OC} - \overrightarrow{OE} &= \frac{1}{2}(\overrightarrow{OG} - \overrightarrow{OE}) \\ \therefore \overrightarrow{OC} &= \frac{1}{2}(\overrightarrow{OE} + \overrightarrow{OG}) \quad \checkmark \\ &= \frac{1}{2}\left(\begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 15 \\ -2 \end{pmatrix}\right) \\ &= \begin{pmatrix} 2 \\ 9 \\ 1 \end{pmatrix}\end{aligned}$$

$$\therefore C(2, 9, 1) \quad \checkmark$$

- ii. ✓ [1] for correct radius with working out
 ✓ [1] for correct vector equation for sphere S

$$\begin{aligned}\overrightarrow{CE} &= \begin{pmatrix} -2 \\ -6 \\ 3 \end{pmatrix} \\ |\overrightarrow{CE}| &= 7\end{aligned}$$

$\therefore$ Sphere S has radius 7 units.

$$\text{Hence the vector equation of sphere } S \text{ is: } \left| \mathbf{y} - \begin{pmatrix} 2 \\ 9 \\ 1 \end{pmatrix} \right| = 7. \quad \checkmark$$

- iii. ✓ [1] for proving equation (3.1)
 ✓ [1] for proving equation (3.2)

$PC \perp CE$ and $PC \perp CF$, since PC is perpendicular to base $EFGH$.

$$\overrightarrow{CP} = \begin{pmatrix} a-2 \\ b-9 \\ c-1 \end{pmatrix}, \quad \overrightarrow{CE} = \begin{pmatrix} -2 \\ -6 \\ 3 \end{pmatrix}, \quad \overrightarrow{CF} = \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}$$

From $PC \perp CE$:

$$\begin{aligned}\begin{pmatrix} a-2 \\ b-9 \\ c-1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -6 \\ 3 \end{pmatrix} &= 0 \\ -2a + 4 - 6b + 54 + 3c - 3 &= 0 \\ \therefore -2a - 6b + 3c &= -55 \quad \checkmark\end{aligned}\tag{3.1}$$

From $PC \perp CF$:

$$\begin{aligned} \begin{pmatrix} a-2 \\ b-9 \\ c-1 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix} &= 0 \\ -6a + 12 + 3b - 27 + 2c - 2 &= 0 \\ \therefore -6a + 3b + 2c &= 17 \quad \checkmark \end{aligned} \tag{3.2}$$

- iv. $\checkmark$ [1] for a correct vector on ℓ
 $\checkmark$ [1] for one correct vector for $\overrightarrow{CP}$
 $\checkmark$ [1] for one correct vector for $\overrightarrow{OP}$
 $\checkmark$ [1] for both correct coordinates of P

Equations (3.1) + $2 \times$ (3.2):

$$\begin{aligned} -14a + 7c &= -21 \\ 2a - c &= 3 \end{aligned} \tag{3.3}$$

Let $a = 1$:

$$\begin{aligned} c &= -1 && \text{(from (3.3))} \\ \therefore b &= \frac{25}{3} && \text{(from (3.2))} \end{aligned}$$

So $M(1, \frac{25}{3}, -1)$ is a point on the line ℓ that is perpendicular to base $EFGH$.

$$\begin{aligned} \overrightarrow{CM} &= \begin{pmatrix} -1 \\ -\frac{2}{3} \\ -2 \end{pmatrix} \quad \checkmark \\ |\overrightarrow{CM}| &= \frac{7}{3} \end{aligned}$$

$|\overrightarrow{CP}| = 7$ and $CP \parallel CM$, since P is on sphere S and also on ℓ :

$$\begin{aligned} \overrightarrow{CP} &= \pm 3 \times \overrightarrow{CM} \quad \checkmark \\ \overrightarrow{CP_1} &= \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix}, \quad \overrightarrow{CP_2} = \begin{pmatrix} -3 \\ -2 \\ -6 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \overrightarrow{OP} &= \overrightarrow{OC} - \overrightarrow{CP} \\ \overrightarrow{OP_1} &= \begin{pmatrix} -1 \\ 7 \\ -5 \end{pmatrix}, \quad \overrightarrow{OP_2} = \begin{pmatrix} 5 \\ 11 \\ 7 \end{pmatrix} \quad \checkmark \end{aligned}$$

Hence P has coordinates $(-1, 7, -5)$ or $(5, 11, 7)$. $\checkmark$

iv. **Alternate marking scheme**

- ✓ [1] for correct $\overrightarrow{OP}$ in terms of one parameter only
- ✓ [1] for both correct values of the parameter
- ✓ [1] for one correct vector for $\overrightarrow{OP}$
- ✓ [1] for both correct coordinates of P

From (iii):

$$-2a - 6b + 3c = -55 \quad (3.1)$$

$$-6a + 3b + 2c = 17 \quad (3.2)$$

Equations (3.1) + 2 × (3.2):

$$-14a + 7c = -21$$

$$c = 2a - 3$$

Equations 2 × (3.1) − 3 × (3.2):

$$14a - 21b = -161$$

$$b = \frac{2}{3}a + \frac{23}{3}$$

$$\therefore \overrightarrow{OP} = \begin{pmatrix} a \\ \frac{2}{3}a + \frac{23}{3} \\ 2a - 3 \end{pmatrix}$$

From (ii):

$$\left| \mathbf{y} - \begin{pmatrix} 2 \\ 9 \\ 1 \end{pmatrix} \right| = 7 \quad (3.5)$$

Since P lies on the surface of the sphere, substitute $\overrightarrow{OP}$ into Equation (3.5):

$$\left| \begin{pmatrix} a - 2 \\ \frac{2}{3}a - \frac{4}{3} \\ 2a - 4 \end{pmatrix} \right| = 7$$

$$\frac{1}{3} \sqrt{(3a - 6)^2 + (2a - 4)^2 + (6a - 12)^2} = 7$$

$$49a^2 - 196a + 196 = 441$$

$$a^2 - 4a - 5 = 0$$

$$(a + 1)(a - 5) = 0$$

$$a = -1 \quad \text{or} \quad a = 5 \quad \checkmark$$

$$\therefore \overrightarrow{OP_1} = \begin{pmatrix} -1 \\ 7 \\ -5 \end{pmatrix} \quad , \quad \overrightarrow{OP_2} = \begin{pmatrix} 5 \\ 11 \\ 7 \end{pmatrix} \quad \checkmark$$

Hence P has coordinates $(-1, 7, -5)$ or $(5, 11, 7)$. $\checkmark$